Dr. Apoorva Upadhyaya

Postdoctoral Researcher & Operational Manager (EU Horizon RECITALS) · L3S Research Center, Leibniz University Hannover, Germany

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Profile

NLP researcher (PhD 2026, magna cum laude) and operational manager of the EU Horizon project RECITALS at the L3S Research Center, Leibniz University Hannover. Research on mechanistic interpretability, safety alignment, and the multilingual and cultural behaviour of LLMs, building on work in social media NLP and human studies of AI interaction. Experience in benchmark and evaluation design, EU project work-package leadership and student supervision.

Current focus: interpretability of LLM responses to controversial queries, including from an *agentic AI perspective*, and how open, German & English foundation models (Soofi Project by German AI Association, funded by the Federal Ministry for Economic Affairs and Energy (BMWE)) handle long-form text, including book narratives, song lyrics, and other *copyright-sensitive material*.

Research Interests

My core interest is understanding and evaluating large language models: their internal mechanisms (sparse-autoencoder analysis and other interpretability methods), how safety alignment holds up across languages and cultures, and how well they handle long, naturalistic text such as books and narratives. A growing strand of this work examines what open, German-trained LLMs such as the SOOFI models reproduce from copyrighted material like book narratives and song lyrics. I am equally interested in the human side: how people interact with language models, and how behavioural studies can inform model evaluation.

Beyond this, I work on computational social science and social media NLP (stance, emotion, morality, toxicity and speech acts), and on federated learning and explainable AI for trustworthy, privacy-preserving language technology.

Positions

Postdoctoral Researcher and Operational Manager, EU Horizon project RECITALS · L3S Research Center, Leibniz University Hannover · 2026 – present

- Operational management of the project's research and delivery, coordinating work packages across partners
- Leading work packages and developing the federated learning and explainable AI components; presented at the 2025 EU-project plenary meeting.
- Coordinate and supervise student teams on AI research topics.

Researcher (TV-L 13, 100%), L3S Research Center, Leibniz University Hannover · Oct 2021 – present

- Research on LLM interpretability and safety, stance detection and social media NLP, and federated learning.
- Contributed to the project of SoMeCliCS (Social Media and Climate Change), a German state project, including pilot studies with school students.

Researcher, Genomics Labs, Life Science & Health Care Research & Innovation Lab, Tata Consultancy Services · Aug 2020 – Jun 2021: Built a tool for parsing lab reports from different instruments; initial work on drug prediction.

System Engineer, Tata Consultancy Services · Nov 2014 – Jul 2018

Education

- **Dr. rer. nat. (PhD)**, Leibniz University Hannover, L3S Research Center, Germany · Apr 2023 – Apr 2026 · Advisors: Prof. Wolfgang Nejdl.

- **Master in Technology (M.Tech)**, Computer Science and Technology, Indian Institute of Technology (IIT Patna), India · Aug 2018 – Jul 2020 · CGPA 9.1, second position in the department.
- **Bachelor in Technology (B.Tech)**, Computer Engineering, YMCA University of Science and Technology, India · 2010 – 2014 · CGPA 8.946, first division.

Current Research Highlights

Evaluating and interpreting language models

- *When Safety Speaks a Language:* mechanistic analysis of how safety behaviour and language identity are entangled in LLMs (EACL Commitment 2027).
- *Culture-Wiki:* benchmark of cultural knowledge recall across cultures (Under Review, ARR October 2026).
- *Finding the Cracks:* Auditing Safety Alignment in LLMs through Interpretable Feature Analysis (Under Review, ARR October 2026).
- *Context-Native Recursive Language* Models for medical QA and evidence retrieval over long documents, evaluated on MIRAGE and R2MED (Under Review, ICLR 2027).

Planned direction: evaluating how open, German-trained LLMs (for example the SOOFI models, a collaborative German and European initiative) respond to prompts built from book narratives, song lyrics and other copyright-sensitive material, with a view to long-range narrative understanding.

Planned direction: mechanistic interpretability of LLM responses to sensitive queries on recent geopolitical conflicts, comparing events before and after a model's training cutoff (refusal, hallucination or reasoning by analogy) and, in an agentic variant with search tools, how retrieved context shifts stance, framing and hedging.

Human-centred studies of language and AI

- Study of how secondary-school students interact with AI to detect moral sentiment in social media (Array Q1, 2026), and pilot studies with students at the IGS List school (Hannover) and an Italian secondary school (2022–2023).
- Interpretable, zero-shot stance detection guided by emotion, morality, toxicity and speech-act signals.

Collaboration and leadership

- Research visits to Sapienza University of Rome (Prof. Roberto Navigli, 2025) and IIT Kharagpur (Prof. Niloy Ganguly, 2026).
- Operational management and technical leadership within the EU Horizon RECITALS project.

Publications Google Scholar

Under review and in preparation

Upadhyaya, A., & Sikdar, S. (2026). When Safety Speaks a Language: A Mechanistic Analysis of Safety-Language Identity Entanglement in LLMs. EACL 2027 Commitment.

Moroni, L., et al. (2026). Culture-Wiki: Benchmarking Different Cultures in Knowledge Recalling Tasks. In preparation for ACL ARR October 2026.

Upadhyaya, A., & Sikdar, S. (2026). Finding the Cracks: Auditing Safety Alignment in LLMs through Interpretable Feature Analysis. ACL ARR October 2026.

Wolter, S., Upadhyaya, A., & Fisichella, M. Context-Native Recursive Language Models for Medical Question Answering and Evidence Retrieval. In preparation.

Journal articles

Upadhyaya, A., Schicchi, D., Taibi, D., & Fisichella, M. (2026). Exploring how students interact with AI for detecting moral sentiments in social media. *Array*, 30, 100941. Elsevier (Q1, IF 5.3).

Upadhyaya, A., Nejdl, W., & Fisichella, M. (2025). Interpretable zero-shot stance detection with proactive content intervention. *Information Processing & Management*, 62(6), 104223 (Q1, IF 8.1).

Upadhyaya, A., Fisichella, M., & Nejdl, W. (2023). Towards sentiment and temporal aided stance detection of climate change tweets. *Information Processing & Management*, 60(4), 103325 (Q1, IF 8.1).

Upadhyaya, A., & Chandra, J. (2022). Spotting flares: The vital signs of the viral spread of tweets made during communal incidents. *ACM Transactions on the Web*, 16(4), 1–28.

Saha, T., Upadhyaya, A., Saha, S., & Bhattacharyya, P. (2021). A multitask multimodal ensemble model for sentiment- and emotion-aided tweet act classification. *IEEE Transactions on Computational Social Systems*, 9(2), 508–517.

Conference and workshop papers

Upadhyaya, A., Nejdl, W., & Fisichella, M. (2026). Towards Transparent Stance Detection: A Zero-Shot Approach Using Implicit and Explicit Interpretability. *AAAI ICWSM 2026*.

Murashilli, V. V., Khatri, K. K., Upadhyaya, A., & Fisichella, M. (2026). Which Layers Matter? A Study of Layer-Selective LoRA in Centralized and Federated Fine-Tuning. *ECML PKDD 2026 Workshop* (to appear).

Heitbreder, J., Jesinghaus, C., Köpp, J., Upadhyaya, A., & Fisichella, M. (2026). MTL-FLoRA: Federated Low-Rank Adaptation for Multi-Task Learning. *ECML PKDD 2026 Workshop* (to appear).

Upadhyaya, A., Nejdl, W., & Fisichella, M. (2025). Enhancing Online Climate Discourse: A Two-Stage Framework for Climate Content Categorization and Moderation. *IJCAI 2025*, 9890–9898.

Ghosh, M., Srivastava, S., Upadhyaya, A., Halder, R., & Chandra, J. (2025). Catalog: Exploiting joint temporal dependencies for enhanced phishing detection on Ethereum. *ACM Web Conference 2025*, 969–977.

Upadhyaya, A., Nejdl, W., & Fisichella, M. (2024). Harnessing Empathy and Ethics for Relevance Detection and Information Categorization in Climate and COVID-19 Tweets. *CIKM 2024*, 4091–4095.

Upadhyaya, A., Fisichella, M., & Nejdl, W. (2023). Toxicity, morality, and speech act guided stance detection. *Findings of EMNLP 2023*, 4464–4478.

Upadhyaya, A., Fisichella, M., & Nejdl, W. (2023). Intensity-Valued Emotions Help Stance Detection of Climate Change Twitter Data. *IJCAI 2023*, 6246–6254.

Upadhyaya, A., Fisichella, M., & Nejdl, W. (2023). A multi-task model for emotion and offensive aided stance detection of climate change tweets. *ACM Web Conference 2023*, 3948–3958.

Upadhyaya, A., Fisichella, M., & Nejdl, W. (2023). A multi-task model for sentiment aided stance detection of climate change tweets. *AAAI ICWSM 2023*, 17, 854–865.

Schicchi, D., Upadhyaya, A., Fisichella, M., & Taibi, D. (2023). Using ChatGPT to Enhance Students' Behavior in Social Media via the Moral Foundation Theory.

Upadhyaya, A., Pfeiffer, C., Astappiev, O., Marenzi, I., Lenzer, S., Nehring, A., & Fisichella, M. (2022). How Learnweb can support science education research on climate change in social media. *MIS4TEL*, 149–154. Springer.

Saha, T., Upadhyaya, A., Saha, S., & Bhattacharyya, P. (2021). Towards sentiment and emotion aided multi-modal speech act classification in Twitter. *NAACL-HLT 2021*, 5727–5737.

Priya, S., Upadhyaya, A., Bhanu, M., Dandapat, S. K., & Chandra, J. (2020). Endea: Ensemble based decoupled adversarial learning for identifying infrastructure damage during disasters. *CIKM 2020*, 1245–1254.

Teaching and Supervision

- Course coordination and supervision of student teams on healthcare AI research topics, Leibniz University Hannover (2026).

- Teaching Assistant, Federated Learning coursework (Winter 2024/25 and 2025/26) and AI seminar (Summer 2025), Leibniz University Hannover.
- Mentor of two Bachelor's theses: Federated Learning of Large Language Models on Agricultural Drone Imagery Using Simulated Flight Planning and Exploring ChatGPT for Drone Applications: Prompts, Dialogues, and Task Adaptability.
- Mentor of Master's thesis (Ongoing): Disentangling language and mathematical reasoning in LLMs.
- Supervision of master's students' research on layer-selective LoRA, multi-task federated LoRA, and recursive language models for medical QA, resulting in joint publications and submissions.

Service, Talks and Visits

Program committees

- WAFL Workshop, ECML PKDD 2026 · Demo Track, ECIR 2026.
- Workshop on Generative AI for Disinformation and Misinformation Detection, WSDM 2025 (also session moderator).
- Workshop on High-performance AI in Education, AIxIA 2023.

Reviewing (2022 – present): EMNLP, ARR, ICLR, ECML PKDD, ICWSM, Information Processing & Management, EPJ Data Science, ACM Computing Surveys and others.

Invited and conference talks

- Recent: AAAI ICWSM 2026, Los Angeles, USA · IIT Kharagpur 2026, India · IJCAI 2025, Montreal, Canada · CIKM 2024.
- Doctoral thesis defence talk (2026).
- Talks at ACM Web Conference, IJCAI, EMNLP, and ICWSM (2023).

Research visits

- IIT Kharagpur, India, with Prof. Niloy Ganguly (2026, EU THIRDWAVE project).
- Sapienza University of Rome, Italy, with Prof. Roberto Navigli (2025).

Honors and Distinctions

2026: Magna cum laude, doctoral thesis and defence
2020: Second position in the M.Tech programme, Computer Science and Engineering, IIT Patna (CGPA 9.1)
2018: GATE, 99.93 percentile (All India Rank 687)
2014: B.Tech with first division (CGPA 8.946), YMCA University of Science and Technology
2010: School topper, CBSE senior secondary (93%) and secondary (95.2%) examinations

Skills

Python and deep learning for NLP; LLM fine-tuning (LoRA, federated), interpretability (sparse autoencoders)
Benchmark and evaluation design; experimental design and statistical analysis
Human-subject pilot studies with students; project and work-package management
Languages: English (fluent) and Hindi (fluent)

References

- Prof. Wolfgang Nejdl (PhD Supervisor), Professor, Scientific Director, L3S Research Center, Leibniz Universität Hannover, Germany, nejdl@l3s.de.
- Prof. Marco Fisichella (Research Collaborator), AI Network ELLIS Member, Research Group Leader, Senior Strategy Officer, L3S Research Center, Leibniz Universität Hannover, Germany, mfisichella@l3s.de.
- Dr. Sandipan Sikdar (Research Collaborator), Junior Prof., L3S Research Center, Leibniz Universität Hannover, Germany, sikdar@l3s.de.
- Prof. Roberto Navigli (Ongoing Collaborator), Professor, Sapienza University of Rome, Italy, navigli@di.uniroma1.it.